

Homework 5

MAE 384

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Q1:

Problem Statement: I am to convert 2 2nd order differential equations of motion into 4 1st order and then plot radius, theta, and trajectory over time. I am given an initial height and velocity, and then for Part C do it for circular orbit and initial velocity of 9000m/s.

Solution: I converted the 2 eqns to 4 1st order eqns. I set each variable to a value and then used the given equations to get a system of equations that the Matlab function ode45 can solve. I also used those equations with the initial conditions and some dynamics equations to find the initial values to put into ode45 to get solutions over the given time span.

a)

$$\ddot{r} = r\dot{\theta}^2 - \frac{Gm_e}{r^2}$$

$$\ddot{\theta} = -\frac{2\dot{r}\dot{\theta}}{r}$$

$$x(1) = r$$

$$x(2) = \dot{r}$$

$$x(3) = \theta$$

$$x(4) = \dot{\theta}$$

$$\dot{x}(1) = x(1)(x(4))^2 - \frac{Gm_e}{(x(1))^2}$$

$$\dot{x}(4) = -\frac{2(x(2))(x(4))}{(x(1))}$$

Initial conditions:

$$I_c = \begin{bmatrix} x_0(1) \\ x_0(2) \\ x_0(3) \\ x_0(4) \end{bmatrix} \rightarrow \begin{matrix} \rightarrow \dot{r}_0 \\ \rightarrow \dot{\theta}_0 \\ \rightarrow \theta_0 \\ \rightarrow \dot{\theta}_0 \end{matrix} \quad t=0$$

$$x_0(1) = r(0) = r_e + h(0) = 6378.14 \text{ km} + 772 \text{ km} = 7150.14 \text{ km} \quad (\text{meters})$$

$$x_0(2) = \dot{r}_0 = 0 \quad \text{due to } V \text{ being } \perp \text{ to Earth @ } t=0$$

$$x_0(3) = \theta_0 = 0$$

$$x_0(4) = \omega_0 = \frac{V_0}{r_0} = \frac{(6700 \text{ m/s})}{(7150.14)(1000)} = 9.3704 \times 10^{-4} \text{ rad/s}$$

$$h_0 = 772 \text{ km}$$

$$V_0 = 6700 \text{ m/s}$$

$$G = 6.672 \times 10^{-11} \text{ m}^3/\text{kg}\cdot\text{s}^2$$

$$M_e = 5.9742 \times 10^{24} \text{ kg}$$

$$r(t) = r_e + h(t)$$

$$r_e = 6378.14 \text{ km}$$

$$\omega = \dot{\theta} = \frac{V}{r}$$

$$\dot{x}(1) = x(2)$$

$$\dot{x}(3) = x(4)$$

Input Equations:

$$\dot{x} = \dot{r} \cos \theta - r \dot{\theta} \sin \theta$$

$$\dot{y} = \dot{r} \sin \theta + r \dot{\theta} \cos \theta$$

clc; clear; close all;

```
% Initial conditions found on paper
```

```
rI = 7150.14*1000;  
rDot = 0;  
thetaI = 0;  
thetaDot = 9.3704e-4;
```

```
%ode45
```

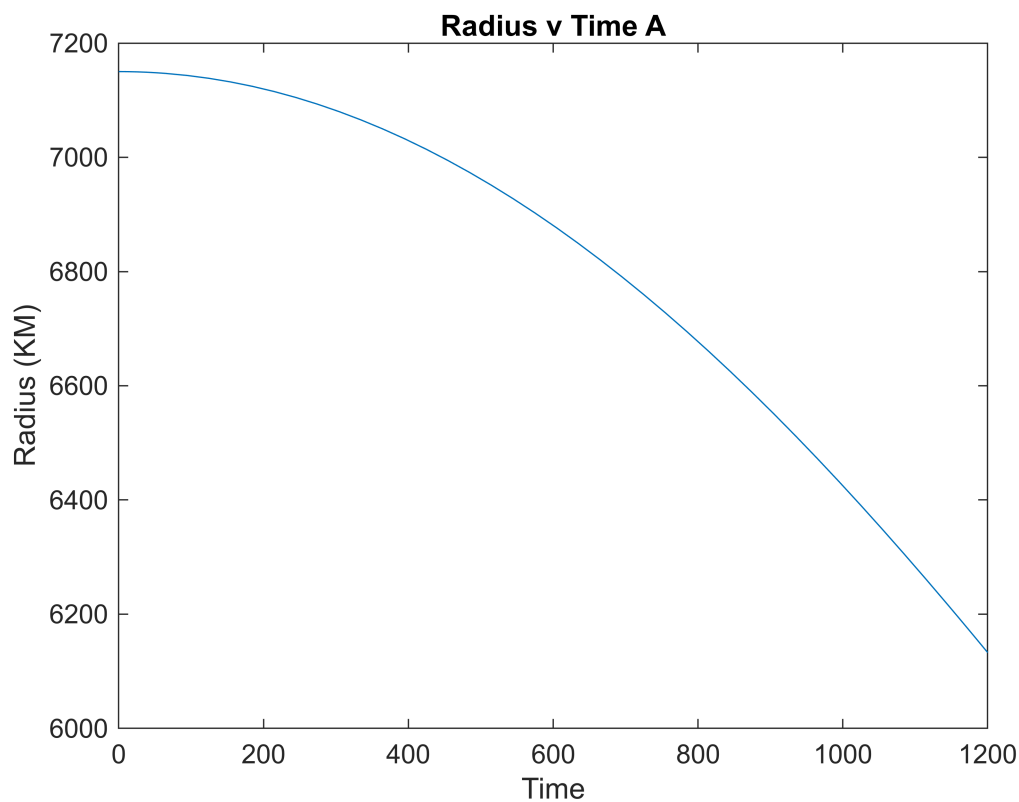
```
[tout, polarout] = ode45(@spacecraftODE, [0 1200], [rI, rDot, thetaI, thetaDot]);
```

```
%Radius and Theta
```

```
rF = polarout(:, 1);  
rFKM = polarout(:, 1)./1000;  
thetaF = polarout(:, 3);
```

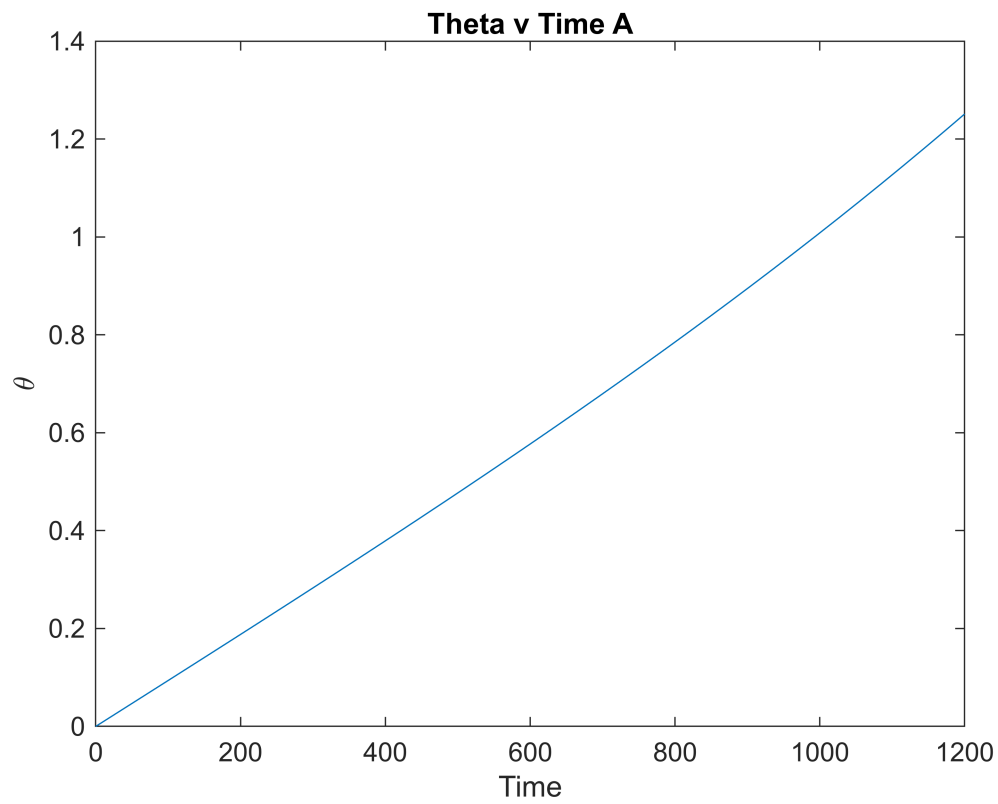
```
%Plots
```

```
figure(1);  
plot(tout, rFKM);  
xlabel('Time');  
ylabel('Radius (KM)');  
title('Radius v Time A');
```

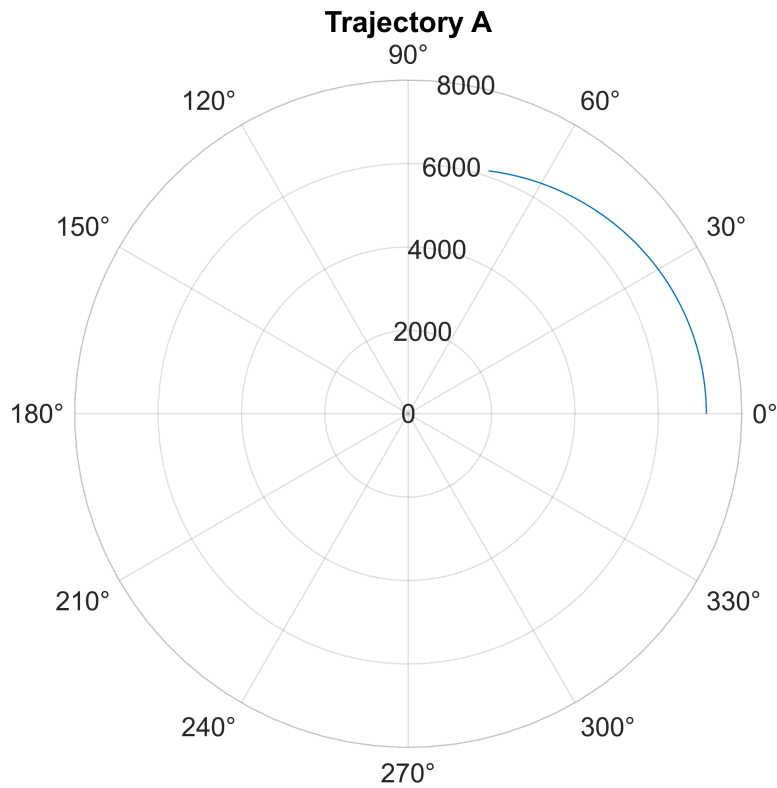


```
figure(2);  
plot(tout, thetaF);  
xlabel('Time');  
ylabel('\theta');
```

```
title('Theta v Time A');
```



```
figure(3);  
polarplot(thetaF, rFKM);  
title('Trajectory A');
```



```
%Values at impact
```

```
[~, impact_index] = min(abs(polarout(:, 1) - 6378.14e3));  
thetaI = polarout(impact_index, 3);
```

```
% Velocities @ Impact
```

```
hvI = polarout(impact_index, 4) * polarout(impact_index, 1);  
vvI = polarout(impact_index, 2);
```

```
fprintf('Theta at impact: %.2f\n', thetaI);
```

```
Theta at impact: 1.04
```

```
fprintf('Horizontal velocity at impact: %.2f m/s\n', hvI);
```

```
Horizontal velocity at impact: 7495.94 m/s
```

```
fprintf('Vertical velocity at impact: %.2f m/s\n', vvI);
```

```
Vertical velocity at impact: -1395.07 m/s
```

Part C:

c) Circular Orbit

$$F_c = F_g$$

$$\frac{mv^2}{R} = \frac{GMm}{R^2}$$

$$v^2 = \frac{GM}{R}$$

$$v = \sqrt{\frac{GM}{R}}$$

% Initial conditions found on paper

Me = 5.9742e24;

G = 6.672e-11;

rI = 7150.14*1000;

vc = sqrt((G*Me)/rI);

rDot = 0;

```

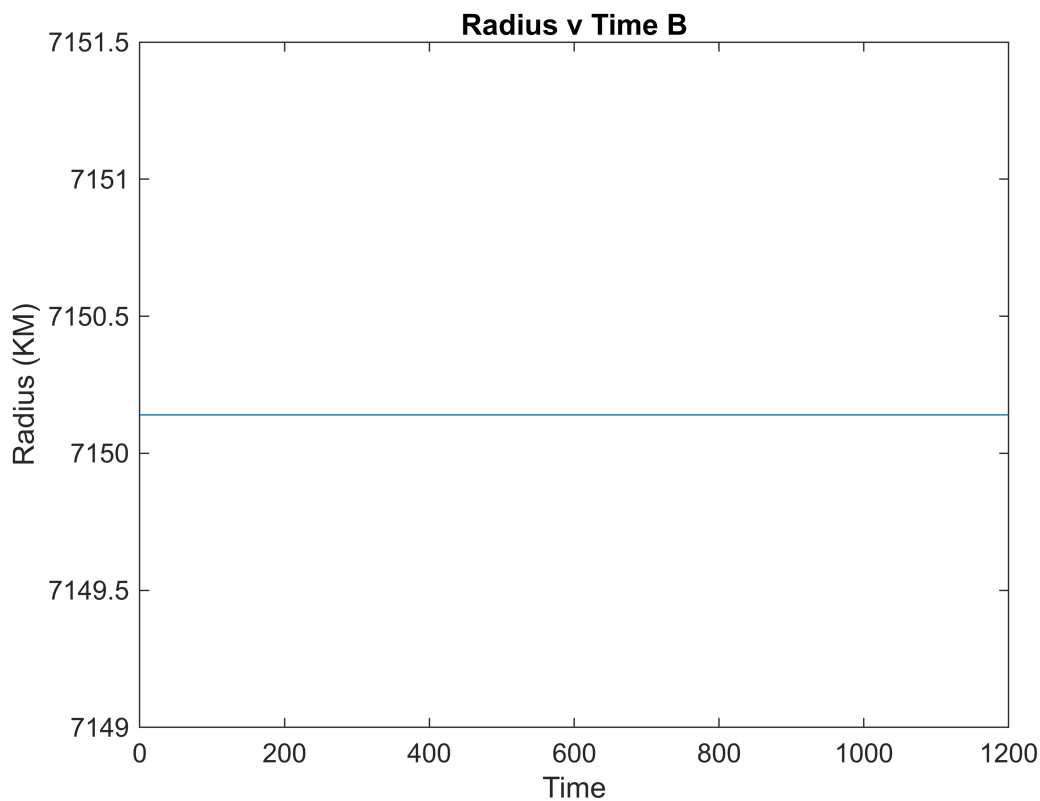
thetaI = 0;
thetaDot = vc / rI;

%ode45
[tout, polarout] = ode45(@spacecraftODE, [0 1200], [rI, rDot, thetaI, thetaDot]);

%Radius and Theta
rF = polarout(:, 1);
rFKM = polarout(:, 1)./1000;
thetaF = polarout(:, 3);

%Plots
figure(1);
plot(tout, rFKM);
xlabel('Time');
ylabel('Radius (KM)');
title('Radius v Time B');

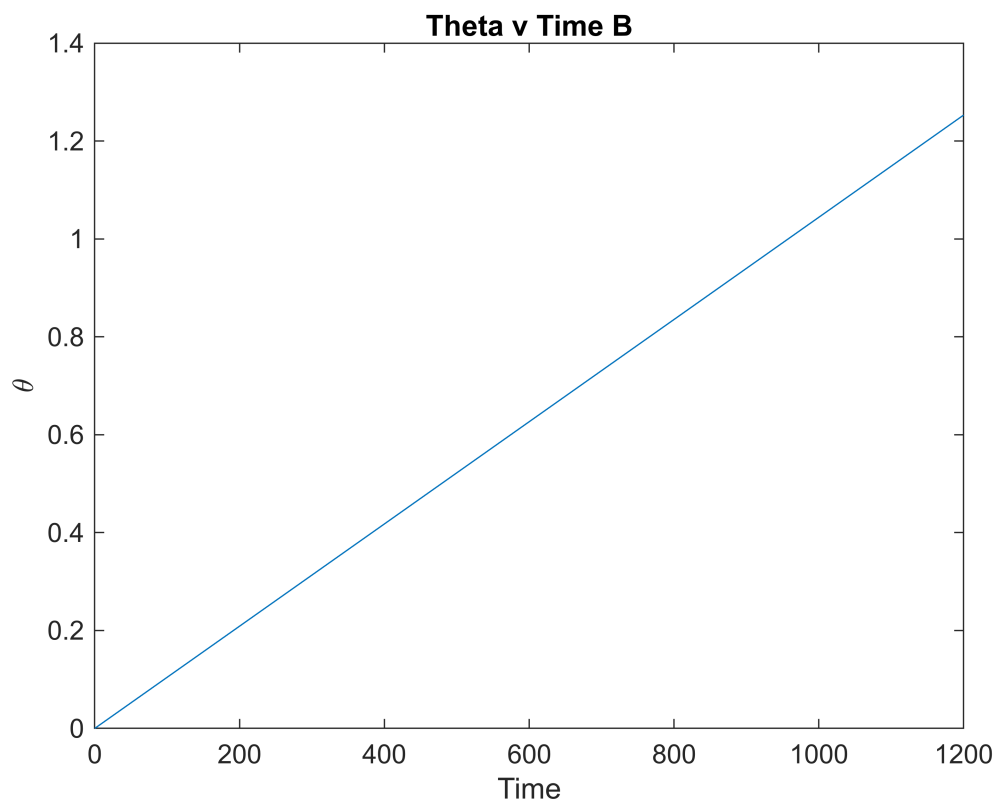
```



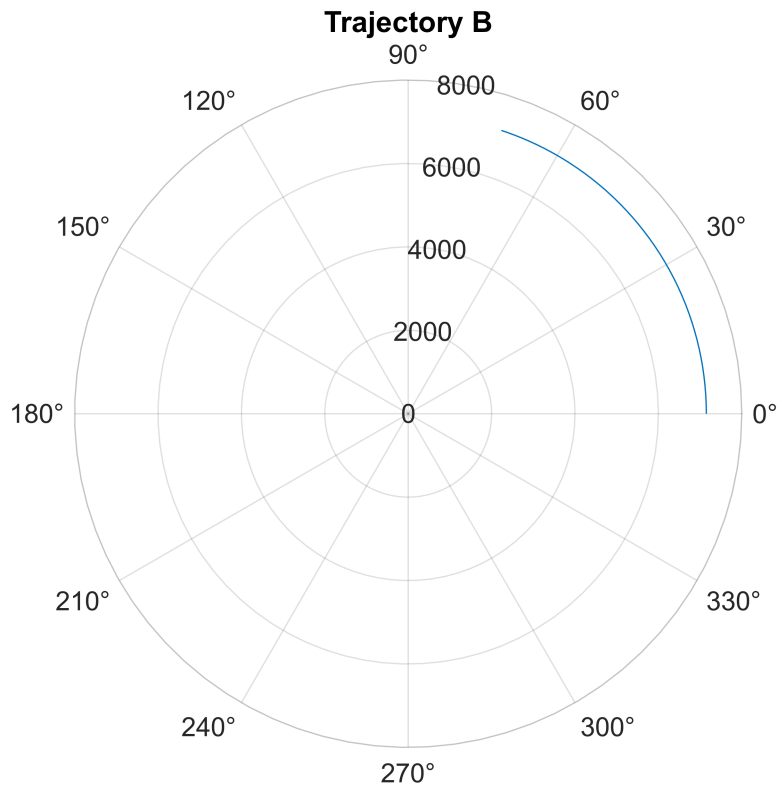
```

figure(2);
plot(tout, thetaF);
xlabel('Time');
ylabel('\theta');
title('Theta v Time B');

```



```
figure(3);  
polarplot(thetaF, rFKM);  
title('Trajectory B');
```

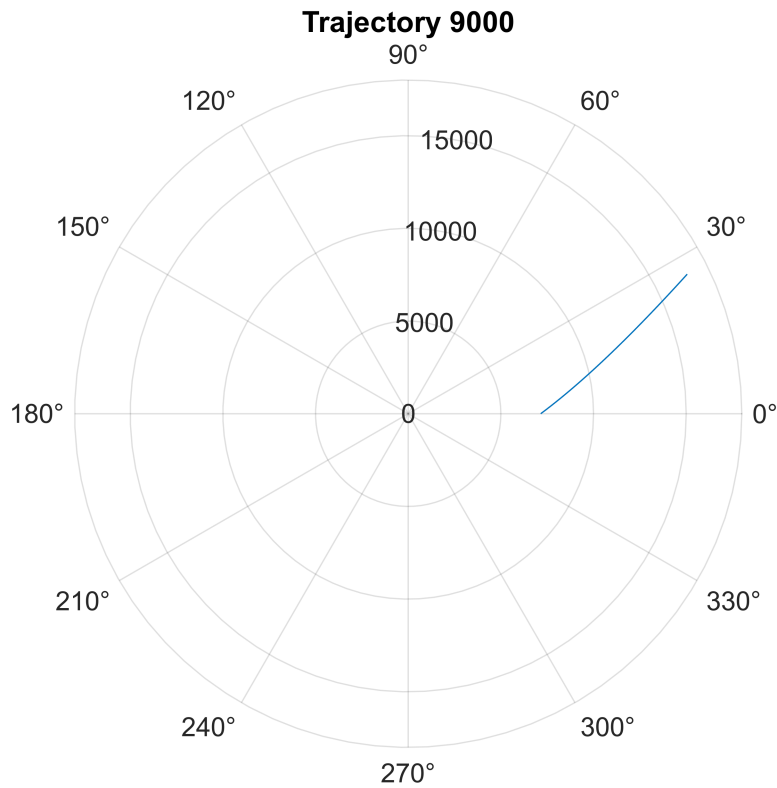
For Initial Velo = 9000:

```
% Initial conditions found on paper
Me = 5.9742e24;
G = 6.672e-11;
rI = 7150.14*1000;
rDot = 9000;
thetaI = 0;
thetaDot = 9.3704e-4;

%ode45
[tout, polarout] = ode45(@spacecraftODE, [0 1200], [rI, rDot, thetaI, thetaDot]);

%Radius and Theta
rF = polarout(:, 1);
rFKM = polarout(:, 1)./1000;
thetaF = polarout(:, 3);

figure(3);
polarplot(thetaF, rFKM);
title('Trajectory 9000');
```



Results: Look above to graphs and displayed values

Discussion: Its clear for the first given values, the spacecraft will fall out of orbit and crash into the planet. While the circular orbit will be a stable well circular orbit. For the 9000m/s initial velocity the spacecraft will achieve a very high trajectory but I believe it is unlikely to have the horizontal velocity needed to maintain an orbit for long.

Appendix: This is the common function code I used in all 3 sections

```
% function polardot = spacecraftODE(t, polarF)
%   %Constants
%   Me = 5.9742e24;
%   G = 6.672e-11;
%
%   %Equations
%   polardot(1) = polarF(2);
%   polardot(2) = polarF(1)*(polarF(4))^2-(G*Me)/((polarF(1))^2);
%   polardot(3) = polarF(4);
%   polardot(4) = -(2*(polarF(2))*(polarF(4)))/(polarF(1));
%
%   %Formatting
%   polardot = polardot';
% end
```

Q2:

Problem Statement: I am given a beam + a partial 4th order ODE of the behavior of a beam being bent with 1 end attached and 1 end free. I am to find the vibration of the beam under different modes and length in a plot.

Solution: I assumed the partial ODE into a solution with 2 different 1st order ODEs multiplied together. I then plugged in the boundary conditions and initial values to get an equation that can be plotted over different n values.

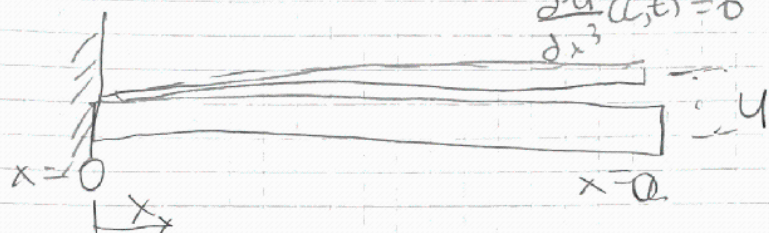
2.
$$\frac{\partial^2 u}{\partial t^2} + \frac{\partial^4 u}{\partial x^4} = 0$$

BC's:

$$u(0, t) = 0$$

$$\frac{\partial u}{\partial x}(0, t) = 0$$

$$\frac{\partial^2 u}{\partial x^2}(L, t) = 0$$

$$\frac{\partial^3 u}{\partial x^3}(L, t) = 0$$


A) Assumed Solution: $u(x, t) = f(x)g(t)$

$$fg'' + f''g = 0 \quad \downarrow \text{divide by } fg$$

$$\frac{g''}{g} + \frac{f''}{f} = 0$$

$$\downarrow$$

$$\frac{g''}{g} = -\frac{f''}{f} = -\lambda^4$$

Spatial:

$$(1) \quad g'' + \lambda^4 g = 0$$

Temporal:

$$(2) \quad f'' - \lambda^4 f = 0 \quad k = \lambda$$

(3) $g(t) = A_1 \cos(\lambda^2 t) + A_2 \sin(\lambda^2 t)$

(4) $f(x) = B_1 \cos(kx) + B_2 \sin(kx)$

b) $f = e^{rx}$
 4th order

$$f^{(4)} = r^4 e^{rx} \rightarrow (2) \quad r^4 e^{rx} - \lambda^4 e^{rx} = 0$$

$$r^4 - \lambda^4 = 0$$

$$(r^2 + \lambda^2)(r^2 - \lambda^2) = 0$$

$$r = \pm \lambda$$

$$r^2 = -\lambda^2$$

$$r = \pm i\lambda$$

c) $x=0$

$$f(0) = D \sin(0) + C \cos(0) + A \cosh(0) + B \sinh(0) = 0$$

$$C + A = 0$$

$$f'(x) = \lambda D \cos(\lambda x) - \lambda C \sin(\lambda x) - \lambda A \sinh(\lambda x) + \lambda B \cosh(\lambda x)$$

$$f'(0) = \lambda D \cos(0) - \lambda C \sin(0) - \lambda A \sinh(0) + \lambda B \cosh(0) = 0$$

$$\lambda D + \lambda B = 0 \rightarrow D + B = 0$$

$$f''(x) = -\lambda^2 D \sin(\lambda x) - \lambda^2 C \cos(\lambda x) - \lambda^2 A \cosh(\lambda x) - \lambda^2 B \sinh(\lambda x)$$

$$x = L$$

$$f''(L) = -(\cos(\lambda L) - \cosh(\lambda L) - D\sinh(\lambda L) - D\cosh(\lambda L)) = 0$$

$$f'''(L) = (\sinh(\lambda L) - \cosh(\lambda L) - D\cos(\lambda L) - D\cos(\lambda L)) = 0$$

$$[(\cos(\lambda L) + \cosh(\lambda L)) (\cos(\lambda L) + \cosh(\lambda L)) + (\sinh(\lambda L) + \cosh(\lambda L)) (\sinh(\lambda L) - \cosh(\lambda L))] = 0$$

$$\cos(\lambda L) \cosh(\lambda L) = 1$$

$$\sinh(\lambda L) - \cosh(\lambda L) = 0$$

$$\boxed{\cos(\lambda L) \cosh(\lambda L) = -1}$$

%Given

LQ2 = 1;

%Function

fun = @(lambda) cosh(lambda*LQ2).*cos(lambda*LQ2) + 1;

% Number of eigenvalues to find

numeigen = 2;

%Bisection Method

to = 1e-6;

maxiter = 100;

lambdamin = 0;

```

lambdamax = 10;
eigenvalues = zeros(numEigen, 1);
eigenvectors = zeros(numEigen, 100);
for i = 1:numEigen
    iter = 0;
    while iter < maxiter
        lambda = (lambdamin + lambdamax)/2;
        equationValue = fun(lambda);
        if abs(equationValue) < to
            break;
        elseif equationValue < 0
            lambdamin = lambda;
        else
            lambdamax = lambda;
        end
        iter = iter + 1;
    end
    eigenvalues(i) = lambda;
    lambdamin = lambdamax;
    lambdamax = lambda + 10;
    x = linspace(0, LQ2, 100);
    eigenvector = sin(lambda*x);
    eigenvectors(i, :) = eigenvector;
end

disp('Eigenvalues:');

```

Eigenvalues:

```
disp(eigenvalues);
```

```

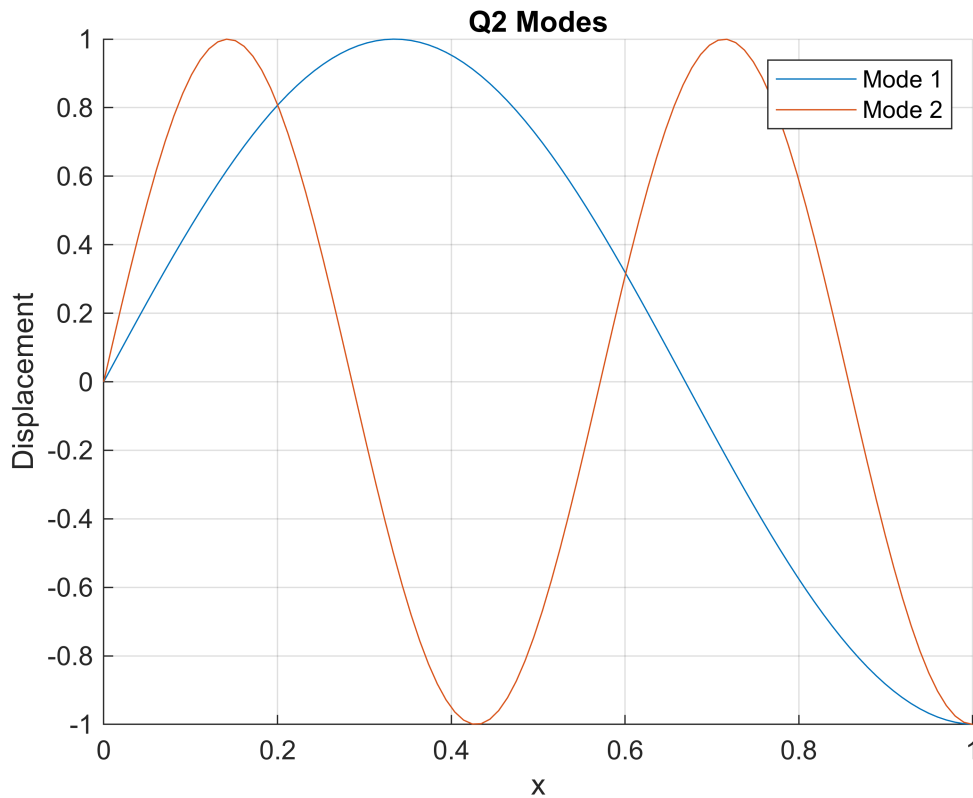
4.6941
10.9955

```

```

%Plot
figure(7);
hold on;
for i = 1:numEigen
    modeShape = eigenvectors(i, :);
    plot(x, modeShape);
end
grid on;
xlabel('x');
ylabel('Displacement');
title('Q2 Modes');
legend('Mode 1', 'Mode 2');
hold off

```



Results: Look at graph above

Discussion: We can use these found eigenvalues and graphs to determine the vibrations and stresses of the beam under different loading conditions to figure out how strong and how flexible to make the beam.

Q3:

Problem Statement: I am given a heat equation of a 1D rod and 2 sets of conditions with a common K value and plot how the heat distributes over the length of the beam.

Solution: 1 set of conditions was homogenous and 1 was non homogenous. I applies the 2 different methods for each conditions that we learned in class to get an equation we can plot on a graph to determine the heat disappation(Is that the right word) over the length of the beam.

PartA:

3

$$a) \quad T(x, 0) = 5 \sin \frac{\pi x}{L}$$

$$T(0, t) = 0$$

$$T(L, t) = 0$$

$$\frac{kt}{L^2} = 0.1$$

$$\frac{\partial T}{\partial t} - k \frac{\partial^2 T}{\partial x^2} = 0$$

$$\text{Assume } T(x, t) = f(x)g(t)$$

$$fg' - k f''g = 0$$

$$\frac{g'}{k_4} = \frac{f''}{f} = -\lambda^2$$

$$f'' + \lambda^2 f = 0$$

$$g' + \lambda^2 k_4 = 0$$

$$f(x) = A_1 \cos \lambda x + A_2 \sin \lambda x$$

$$g(t) = B e^{-\lambda^2 k t}$$

$$\text{BC: } F(0) = A_1 \cos(0) + A_2 \sin(0) = 0$$

$$A_1 = 0$$

$$F(L) = A_1 \cos(\lambda L) + A_2 \sin(\lambda L) = 0$$

$$\lambda = n\pi$$

$$\lambda = \frac{n\pi}{L}$$

$$T(x, t) = [A_2 \sin(\lambda L)] [B e^{-\lambda^2 k t}]$$

IC's:

$$T(x,t) = [A_2 \sin(\lambda x)] [B e^{-\lambda^2 k t}]$$

$$T(x,0) = \sum_{n=1}^{\infty} (c_n \sin(\frac{n\pi x}{L})) = 5 \sin(\frac{\pi x}{L})$$

$n=1$

$$c_n = 5$$

$$T(x,t) = \sum_{n=1}^{\infty} 5 e^{-(\frac{n\pi}{L})^2 k t} \sin(\frac{n\pi x}{L})$$

B)

$$T(x,0) = 0$$

$$T(0,t) = 2$$

$$T(L,t) = 2$$

$$\text{Assume } T(x,t) = f(x)g(t)$$

$$f_y' - k f'' = 0$$

$$\frac{g'}{k g} = \frac{f''}{f} = -\lambda^2$$

$$f(x) = A_1 \cos \lambda x + A_2 \sin \lambda x$$

$$g(t) = 0 e^{-\lambda^2 k t}$$

$$f(0) = 2$$

$$f(L) = 2$$

$$f(x) = 2 \cos \lambda x$$

$$T(x,t) = 2 \cos \lambda x e^{-\lambda^2 k t}$$

```
LQ3A = 1;
x = linspace(0,LQ3A,1000);
t = 0.1;
```

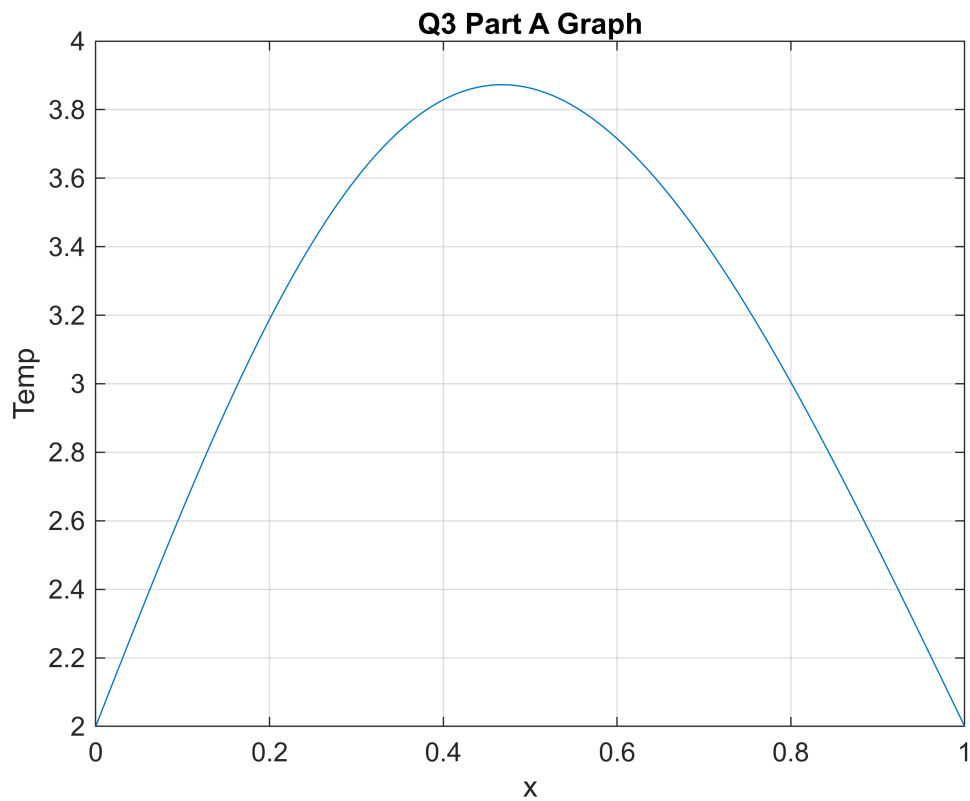
```

N = 100;
T = zeros(size(x));

for n = 1:N
    T = T + 5*exp(-(n*pi)^2*0.1)*sin((n*pi*x)/LQ3A);
end

T = T + 2;
figure(8);
plot(x,T);
hold on
grid on;
xlabel('x');
ylabel('Temp');
title('Q3 Part A Graph');
hold off

```



Part B:

ICs:

$$T(x,t) = [A_2 \sin(\lambda x)] [B e^{-\lambda^2 k t}]$$

$$T(x,0) = \sum_{n=1}^{\infty} (c_n \sin(\frac{n\pi x}{L})) = 5 \sin(\frac{\pi x}{L})$$

$$n=1$$

$$c_n = 5$$

$$T(x,t) = \sum_{n=1}^{\infty} 5 e^{-(\frac{n\pi}{L})^2 k t} \sin(\frac{n\pi x}{L})$$

B)

$$T(x,0) = 0$$

$$T(0,t) = 2$$

$$T(L,t) = 2$$

Assume $T(x,t) = f(x)g(t)$

$$f g' - k f'' g = 0$$

$$\frac{g'}{k g} = \frac{f''}{f} = -\lambda^2$$

$$f(x) = A_1 \cos \lambda x + A_2 \sin \lambda x$$

$$g(t) = B e^{-\lambda^2 k t}$$

$$T(x,t) = \sum_{n=1}^{\infty} \frac{2}{L} \sin(\frac{n\pi x}{L}) e^{-\lambda^2 k t}$$

BC:

$$f(x) = A_1 \cos(x) + A_2 \sin(x) = 2$$

$$A_2 = 2$$

$$f(L) = A_1 \cos(\lambda L) + 2 \sin(\lambda L) = 2$$

$$\lambda L = n\pi$$

$$\lambda = \frac{n\pi}{L}$$

$$A_1 \cos\left(\frac{n\pi}{L}\right) + 2 \sin\left(\frac{n\pi}{L}\right) = 2$$

$$\text{Let } T(x,t) = \psi(x) + \phi(x,t) \quad \text{non homogeneous}$$

$$\psi(0) = \bar{T}_0 \quad \psi(L) = \bar{T}_L$$

$$\phi(0,t) = 0 \quad \phi(L,t) = 0$$

$$\frac{\partial \phi}{\partial t} - k \left[\frac{\partial^2 \psi}{\partial x^2} + \frac{\partial^2 \phi}{\partial x^2} \right] = 0$$

$$k \frac{\partial^2 \psi}{\partial x^2} = 0$$

$$\frac{\partial \phi}{\partial t} - k \frac{\partial^2 \phi}{\partial x^2} = 0$$

$$\frac{\partial^2 \psi}{\partial x^2} = 0$$

$$\frac{\partial^2 \psi}{\partial x^2} = C_1$$

$$\psi(x) = C_1 x + C_2$$

$$\psi(L) = \bar{T}_L = C_1 L + \bar{T}_0$$

$$\psi(0) = \bar{T}_0 = C_2$$

$$C_1 = \frac{\bar{T}_L - \bar{T}_0}{L}$$

$$T(x, t) = \frac{T_L - T_0}{L} + T_0 + \sum_{n=1}^{\infty} C_n e^{-n^2 \pi^2 k t} \sin(n \pi x)$$

IC: $T(0, t) = 2$ $T(L, t) = 2$

$$\frac{2-2}{L} + 2 + \sum_{n=1}^{\infty} C_n e^{-n^2 \pi^2 k t} \sin(n \pi x)$$

$$C_n = 2 \left[\frac{-1}{n\pi} (T_i - T_0) (\cos(n\pi)) - 1 \right]$$

$$T(x, t) = 2 + \sum_{n=1}^{\infty} \left[\frac{-2}{n\pi} (-2) \cos(n\pi) - 1 \right] e^{-\frac{k t n^2 \pi^2}{L}}$$

$$\sin\left(\frac{n\pi x}{L}\right)$$

```
LQ3B = 1;
x = linspace(0, LQ3B, 1000);
```

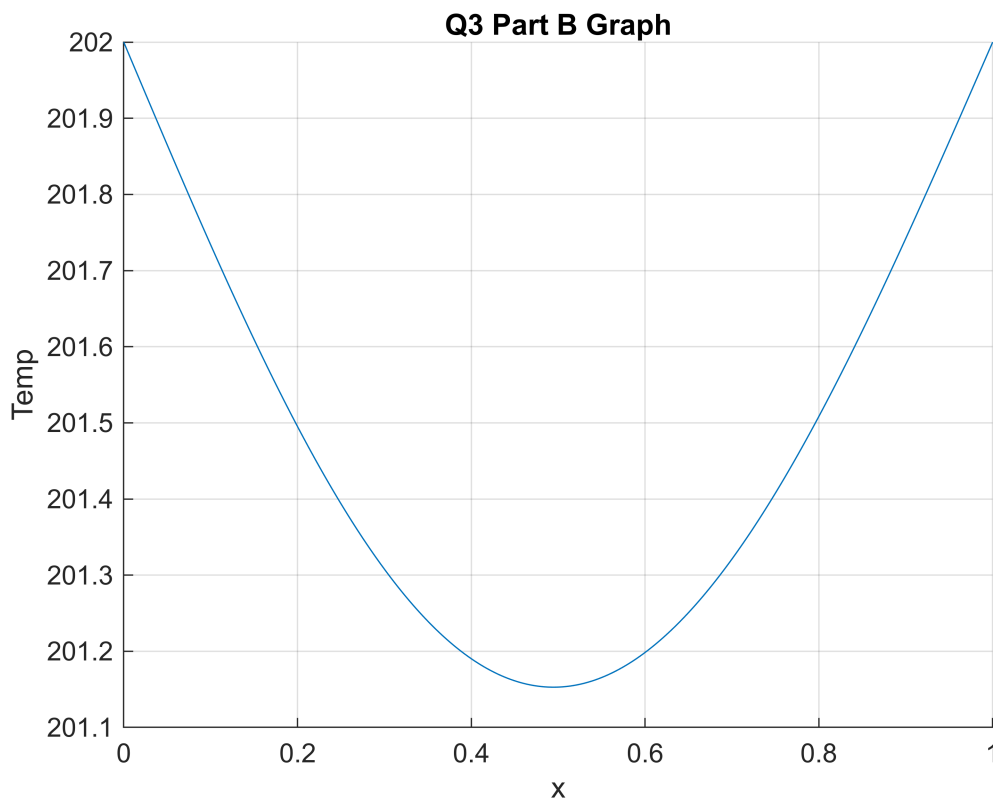
```

t = 0.1;
N = 100;
T = zeros(size(x));

for n = 1:N
    T = 2 + (T + ((-2)/(n*pi))*(-2)*cos(n*pi)-1)*exp(-0.1*n^2*pi^2)*sin((n*pi*x)/LQ3B));
end

T = T + 2;
figure(9);
hold on
plot(x,T);
grid on;
xlabel('x');
ylabel('Temp');
title('Q3 Part B Graph');
hold off

```



Results: Look at graphs above

Discussion: For part A, the heat applies to the center of the beam and dissapates over the center, the ends are not heated. For part A, it is applied at the end of the beam and dissapates near the end while the center is barely heated. We can use this equation for other objects to see how heat disappates for design purposes.